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Multi-Granularity Ensemble Interaction Graph Modeling for Knowledge Tracing

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ABSTRACT

Knowledge tracing (KT) is a crucial educational tool that models students' mastery of various knowledge concepts by analyzing their historical learning records. Mainstream studies have turned to Deep Neural Networks (DNNs) to effectively trace students' knowledge states. Nonetheless, these methods may encounter performance barriers due to two critical constraints: the treatment of learning records as singular time series data and the absence of accounting for interaction correlations. To overcome the limitations, we introduce a novel KT model, named **M**ulti-**G**ranularity Ensemble **I**nteraction **G**raph Modeling for **K**nowledge **T**racing (MGIGKT). Our model incorporates two modules designed to model students' knowledge states from the perspectives of temporal perception and interaction dependency, respectively. This approach allows for a more comprehensive understanding of students' learning dynamics and the correlations between their interactions. We have conducted extensive experiments on four datasets to evaluate the superiority and effectiveness of our proposed MGIGKT model. The results of these experiments provide empirical evidence of the model's enhanced ability to trace students' knowledge states accurately.

1. Introduction

In recent years, the demand for intelligent educational services among students has surged [1–3]. Consequently, online education systems have become increasingly vital in addressing this demand. One such critical component that has come to the forefront is KT, which provides students with immediate and precise feedback. KT entails examining students' historical learning records to evaluate their competency in different knowledge areas, with the goal of forecasting their academic performance ahead. This not only allows for the delivery of varied learning resources but also permits the tailoring of individualized learning trajectories to suit personal educational needs. Moreover, KT underpins adaptive education, streamlining the personalization of educational content and its complexity in accordance with students' learning advancements and knowledge levels. This encompasses features such as automated test creation and post-lesson tasks, all geared towards a more focused and impactful learning journey. By harnessing the power of KT through the analysis of students' historical learning records, online education systems can acquire in-depth insights into their educational states, significantly enhancing the online learning experience for both students and educators.

Currently, numerous studies have been dedicated to exploring the task of KT [4–7]. Due to the powerful feature extraction capabilities of DNNs for modeling complex cognitive processes, recent researches have predominantly relied on more complex and accurate DNNs architectures. These methods can be broadly categorized into two types: conventional KT methods and graph-based KT methods. The first category of conventional KT methods models students' knowledge states from a temporal perspective using deep learning techniques, aiming to dynamically capture the evolving proficiency of students. For example, DKT [8] is the first work to apply Recurrent Neural Networks (RNNs) to the KT task. Due to its simple structure and efficient performance, it has been widely adopted as a paradigm by subsequent deep learning KT models. Following this, more types of neural network technologies have been introduced to improve temporal KT models. For example, DKVMN [9] introduces a memory-augmented network that uses memory networks to store and update knowledge concepts and students' knowledge states. Meanwhile, the SAKT [10] model incorporates an attention mechanism, which calculates the correlation between the exercise to be predicted and the historical interactions in the student's

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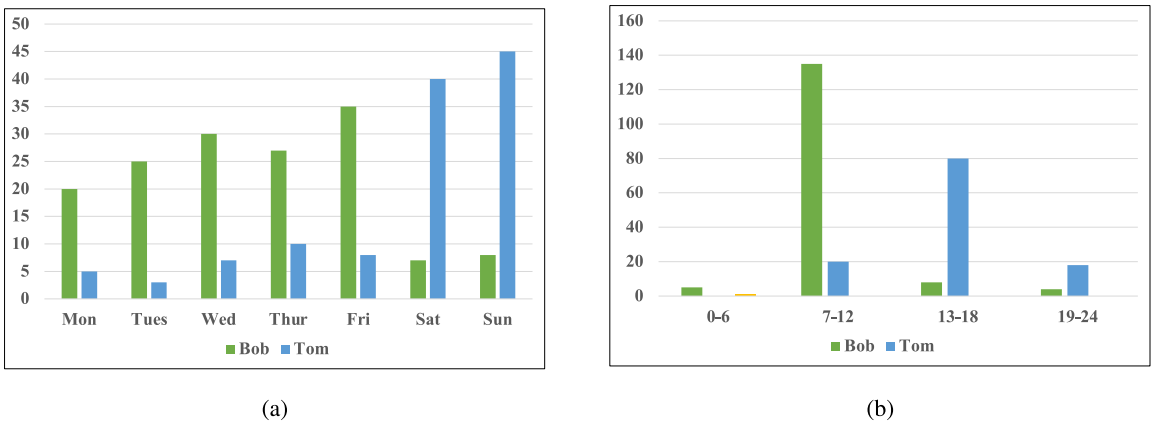


Fig. 1. Distribution histograms of learning records of different students within a week at different granularities: (a) per day; (b) per hour.

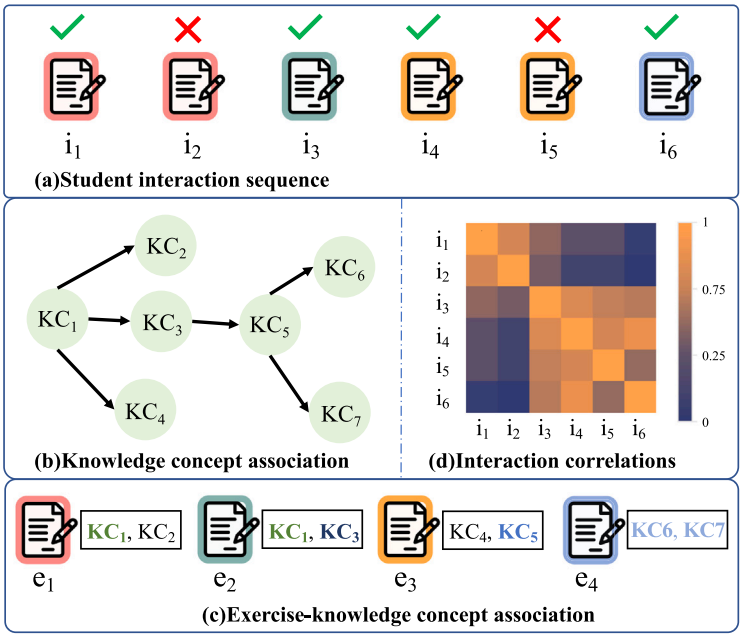


Fig. 2. Example of interaction correlation. (a) illustrates the student's historical interaction sequence. (b) shows the sequential relationships among knowledge concepts. (c) depicts cases where exercises share the same knowledge concept and the knowledge concepts involved have a sequential relationship. (d) presents the final interaction correlations calculated using Eq. (9) mentioned in Section 3.2.3.

learning sequence, thereby capturing changes in the student's knowledge state more accurately. These methods primarily focus on the temporal relationships within learning sequences. Recently, recognizing that knowledge concepts are not completely independent but have latent correlations, and considering the rapid theoretical and practical development of Graph Neural Networks (GNNs) in recent years, the second category of graph-based KT methods has emerged. For example, GKT [11] treats students, exercises, and knowledge concepts as nodes in a graph, with interactions and correlations between knowledge concepts as edges. It uses a GNN to model students' knowledge states, capturing these potential correlations and improving the predictive accuracy of the model.

Despite the significant success of existing KT methods, two major limitations still need to be addressed: (1) *Single granularity modeling of interaction time information*: Current KT models often model the dynamic changes in students' knowledge states at only a single granularity. However, in reality, students' learning records usually exhibit different distribution patterns at various granularities. As shown in Fig. 1, we present the distribution histograms of the learning records of two students. It can be observed that different students often have different learning patterns. For example, Bob prefers to study on weekday mornings, while Tom prefers to study extensively on weekend afternoons. The complexity of these learning patterns poses a challenge to single-granularity modeling of existing methods. Therefore, fully exploring the different learning patterns exhibited by students at multiple granularities is crucial for the success of KT models. (2) *Insufficient capture of latent correlations between interactions*: To achieve good predictions for the next exercise, studies have shown that considering the correlations between historical interactions is highly beneficial [11–13]. Existing models fail to fully uncover the deep connections between historical interactions, such as knowledge concept relationships between exercises. More specifically, these inherent correlations include shared knowledge concepts between exercises and the sequential relationships between the contained knowledge concepts. Meanwhile, students' interaction sequence not only include the exercises but also their responses to these exercises. Therefore, when evaluating the correlation between interactions, it is essential to consider both the knowledge concept relationships and the correctness of the responses. To illustrate the importance of knowledge concept relationships and the correctness of the responses in calculating interaction correlations, we present in Fig. 2(a)

an interaction sequence of a student over six historical timesteps. This sequence includes the exercises the student interacted with and their responses at each timestep. Fig. 2(b), (c), and (d) respectively depict the sequential relationships among knowledge concepts, cases where exercises share the same knowledge concept and the knowledge concepts involved have a sequential relationship, and the final interaction correlations calculated using Eq. (9) mentioned in Section 3.2.3. In our example, from the perspective of shared knowledge concepts, although i_1, i_2 and i_3 all share knowledge concept KC_1, the correlation between i_1 and i_3 is higher since the student's responses to i_1 and i_3 are consistent, while the response to i_2 differs. Similarly, from the perspective of sequential knowledge concepts, interactions i_4, i_5 and i_6 involve the sequential knowledge concepts KC_5, KC_6 and KC_7, further indicating high correlation among these exercises. Notably, since the responses for i_4 and i_6 are consistent, their correlation is higher than that between i_5 and i_6. Therefore, considering both the latent correlations in the student's historical interactions and their responses on the exercises helps achieve more accurate interaction correlations, which is crucial for enhancing the performance of the KT model.

In this paper, we propose a model named MGIGKT, designed to explore students' learning patterns at multiple granularities and comprehensively model the latent correlations in student interactions, aiming to achieve a more accurate representation of students' knowledge states. Specifically, on one hand, to model learning patterns at multiple granularities, we introduce a *Multi-Scale Sequence Perception Modeling module* to encode students' historical learning records at different time scales. To reduce the computational complexity of encoding, we incorporate an L2-regularized Transformer to improve efficiency. Through a multi-head attention mechanism, we can obtain a knowledge state that integrates multiple scales. On the other hand, to comprehensively model the latent correlations between interactions, we propose an *Interaction Graph Dependency Modeling module*, which creates interaction graphs from the perspectives of similarity and dissimilarity. We design a dual-channel encoding mechanism to understand students' knowledge states from these perspectives. Additionally, we employ cross-channel contrastive learning to optimize the learned knowledge states, ensuring that the knowledge states of the same student remain stable across different perspectives. Finally, the *Fusion and Prediction module* fully considers the informative knowledge states modeled by the previous two modules to perform accurate and comprehensive KT.

To summarize, the main contributions of our work are as follows:

- **Model learning patterns at multiple granularities:** We find that different students exhibit varying learning patterns at different time granularities. To address this, we propose the Multi-Scale Sequence Perception Modeling module to comprehensively model students' diverse learning patterns across multiple granularities, resulting in more accurate knowledge states.
- **Model the latent correlations between interactions:** We discover more complex latent correlations among students' historical interactions. To tackle this, we introduce the Interaction Graph Dependency Modeling module, which constructs interaction graphs and utilizes a dual-channel encoding component with a cross-channel contrastive learning mechanism to delve into the underlying correlations in student interactions.
- **Experimental evaluation results:** Extensive experiments conducted on four KT datasets validate the effectiveness of MGIGKT in assessing students' knowledge states.

The remainder of this paper is structured as follows: In Section 2, we provide a comprehensive review of related works pertaining to our research problem. Detailed information about the overall pipeline and components of our model is illustrated in Section 3. In Section 4, we analyze the experimental results obtained from our research. Finally, in Section 5, we present the concluding remarks and summarize our work.

2. Related work

In this section, we will introduce relevant work on KT categorized into two types: conventional KT methods and graph-based KT methods.

2.1. Conventional KT methods

In recent years, with the advancement of deep learning, an increasing number of DNNs models have been applied in the field of KT [14–17]. DKT [8], as a pioneering work, is the first KT model to use RNNs for sequential modeling of students' historical learning records. It achieves better performance than traditional Bayesian KT. DKVMN [9] introduces a dynamic key–value memory network to store and update students' knowledge states. ATKT [18], proposed by Guo et al. combines LSTM and attention networks to update students' knowledge states. Meanwhile, to address the overfitting issue presented in existing works, this study introduces adversarial training, training the model with adversarial samples and original samples simultaneously to enhance the model's robustness. MRTKT [19] introduces an attention mechanism, considering complex cross effects in modeling student knowledge states, such as adjacent interaction, target-dependent interaction, and non-adjacent interaction in students' historical learning records. The CORE [20] introduces causal reasoning in KT to mitigate the bias issue stemming from the imbalance in the distribution of correct and incorrect answers to exercises. All these approaches model students' historical learning records as time-series data but overlook the latent correlations between students' interactions.

2.2. Graph-based KT methods

However, in the reality of learning experiences, students' historical learning records are not mutually independent; due to the correlation of knowledge concepts, there exists latent interdependence between interactions [21,22]. Consequently, many studies focus on leveraging interaction graphs to enhance the performance of KT models. GKT [11] is the first to model knowledge concepts as a graph based on their correlations and utilizes Graph Convolutional Networks (GCNs) to model and update students' knowledge states. DGEKT [12] categorizes interactions into two types of hypergraphs based on whether the knowledge concepts involved in the exercises are the same. Simultaneously, it constructs directed transition graphs based on the temporal sequence of interactions, using knowledge distillation networks to improve the model's accuracy. HGKT [13] models dependency relationships in the learning process of exercises by constructing hierarchical exercise graphs. However, these graph-based methods construct exercises or interactions as graphs and base subsequent KT processes on these graphs without considering changes in student knowledge states from both static and dynamic perspectives.

In summary, most existing KT methods mainly model students' knowledge states either statically or dynamically, or they perform temporal modeling based on constructed graphs. However, there is a lack of consideration for simultaneously modeling students' knowledge proficiency from both static and dynamic perspectives.

3. Method

In this section, we first formalize the KT problem, followed by a detailed description of our method MGIGKT.

Table 1
Key notations and explanations of MGIGKT framework.

Notations	Explanation
$\mathcal{S}$	A set of students
$\mathcal{E}$	A set of exercises
$\mathcal{KC}$	A set of knowledge concepts
$\mathbf{H}^S$	Students knowledge state modeled from a dynamic perspective
$\mathbf{A}^{(sm)}/\mathbf{A}^{(ns)}$	The adjacency weight matrix of similar/dissimilarity interaction graphs
$\mathbf{H}^G$	Students knowledge state modeled from a static perspective
$\mathbf{h}_t$	The ultimate knowledge state at the timestep t
$y_{t+1}, \hat{y}_{t+1}$	The actual response and predicted probability at timestep $t+1$
$\mathbf{W}$	All references to '$\mathbf{W}$' in this paper are related to weight parameter matrices

3.1. Problem statement

In the KT task, let $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$, $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$, and $\mathcal{KC} = \{kc_1, kc_2, \ldots, kc_K\}$ denote a set of N students, a set of M exercises and a set of K knowledge concepts, respectively. Suppose a student s's historical learning records of length t is represented as $\mathcal{I}_s = \{(e_1, \mathcal{KC}_1, a_1), (e_2, \mathcal{KC}_2, a_2), \ldots, (e_t, \mathcal{KC}_t, a_t)\}$. Each interaction is represented by a triple, taking the interaction at timestep t as an example, where e_t ($e_t \in \mathcal{E}$) denotes the exercise answered by the student at the timestep t, $\mathcal{KC}_t \subset \mathcal{KC}$ is the subset of knowledge concepts contained in $e_t, a_t \in \{0, 1\}$ is the correctness of responses, typically, $a_t = 1$ if the student answers e_t correctly, otherwise $a_t = 0$. To provide a clear illustration of the model structure, we present a summary of the key mathematical notations in Table 1. These notations will be utilized throughout the paper. Specifically, we use bold uppercase letters to represent matrices, bold lowercase letters to represent vectors, and calligraphy uppercase letters to represent sets.

KT Task Definition: Given the learning records $\mathcal{I}_s$ of the student s up to timestep t, our objective is to evaluate the mastery of the student on various knowledge concepts after answering each exercise. This is achieved by predicting the probability of students correctly answering a new exercise e_{t+1} at timestep $t+1$, denoted as $\hat{y}_{t+1} = p(a_{t+1} = 1 | e_{t+1}, \mathcal{I}_s)$, and to trace students' knowledge states.

3.2. MGIGKT

In this section, we will introduce the overall process of the proposed MGIGKT model and provide detailed explanations of each component. The architecture of the MGIGKT model is illustrated in Fig. 3. The primary aim of MGIGKT is to model the knowledge states of students from both temporal and interaction graph perspectives, representing dynamic and static viewpoints, respectively. To achieve this goal, we design three modules corresponding to the aforementioned modeling processes. These modules are the *Multi-Scale Sequential Perception Modeling module*, *Interaction Graph Dependency Modeling module*, and *Fusion and Prediction module*. The *Multi-Scale Sequential Perception Modeling module* primarily involves granular partitioning and encoding of historical learning records to obtain multi-granularity knowledge states of the student over time. The *Interaction Graph Dependency Modeling module* introduces interaction graph filters, which use low-pass and high-pass filtering based on similar and dissimilar interaction graphs, respectively, to derive the student's static knowledge state. Finally, the *Fusion and Prediction module* integrates the dynamic and static knowledge states. We utilize the final knowledge state to predict the student's response in the next timestep.

3.2.1. Embedding layer

We first initialize embedding matrices for questions, knowledge concepts, and response results as $\mathbf{Q} \in \mathbb{R}^{M \times d}$, $\mathbf{KC} \in \mathbb{R}^{K \times d}$, and $\mathbf{A} \in \mathbb{R}^{2 \times d}$, respectively, where d represents the embedding dimension. To obtain the representation of the historical learning records of student s, we look up the aforementioned three matrices to obtain the exercise embedding matrix $\mathbf{E}^s$ and the interaction embedding matrix $\mathbf{I}^s$.

$$\mathbf{E}_i^s = \mathbf{q}_i \oplus \left(\frac{1}{m} \sum_{kc_k \in \mathcal{KC}_i} \mathbf{kc}_k\right), \tag{1}$$

$$\mathbf{I}_i^s = \mathbf{q}_i \oplus \left(\frac{1}{m} \sum_{kc_k \in \mathcal{KC}_i} \mathbf{kc}_k\right) \oplus \mathbf{a}_i, \tag{2}$$

where $\oplus$ represents concatenation operation. $\frac{1}{m}\sum_{kc_j \in \mathcal{KC}_i} \mathbf{kc}_j$ denotes the average pooling of knowledge concepts embeddings contained in question q_i.

3.2.2. Multi-scale sequential perception modeling(MSPM)

During the learning process, educational systems, school policies, and students' learning habits determine the establishment of learning cycles, such as teaching months, semesters, and years, as well as various time periods within a day. As a result, students' knowledge states exhibit multi-scale patterns of variation over time, reflecting their staged learning progress across different cycles (e.g., months, semesters, and years). To effectively model this characteristic, we develop a multi-scale Transformer knowledge state encoder with L2 regularization. This encoder is designed to capture features in students' learning sequences simultaneously at both coarse-grained and fine-grained time scales. It can identify and encode students' learning patterns and knowledge mastery levels over different time periods, thereby providing a more comprehensive description of the students' knowledge states.

Coarse-grained modeling of students' knowledge states: The primary purpose of coarse-grained modeling is to capture long-term dependencies in students' historical interaction sequences. In many fields that study sequence-related issues, the Transformer model is widely used for modeling long-term dependencies because it can explore relationships within the sequence through its self-attention mechanism [23–25]. Based on this observation, we employ the Transformer to model sequence relationships. However, in the KT problem, students' learning records can be very long, making the Transformer with its traditional attention mechanism and $O(N^2)$ time complexity for computing self-attention impractical. To mitigate this issue, inspired by previous research [26], we introduce the L2 regularization technique, changing the computation order of the attention mechanism by separately normalizing the rows and columns of the Query and Key matrices, and then activating them using an activation function. These modifications reduce the complexity from $O(N^2)$ to $O(N)$, while still preserving the properties of the attention mechanism and providing sparsity.

Specifically, we first compute $\mathbf{K}^\top \mathbf{V}$ to reduce computational complexity without compromising the learning ability of the attention mechanism. Formally, we rewrite the attention mechanism as follows:

$$\mathbf{H}^E = \rho_1(elu(\mathbf{I}^s\mathbf{W}^Q))(\rho_2(elu(\mathbf{I}^s\mathbf{W}^K))^\top \mathbf{I}^s\mathbf{W}^V), \tag{3}$$

where ρ_1 and ρ_2 are two different mappings. We use row-wise and column-wise L2 normalization to perform these mappings and replace the softmax function. The specific formulas are as follows:

$$\rho_1(\mathbf{Q}_i) = \frac{1}{\sqrt{d}\,\|\mathbf{Q}_i\|_2}\mathbf{Q}_i, \tag{4}$$

$$\rho_2(\mathbf{K}_j) = \frac{1}{\sqrt{N}\,\left\|\mathbf{K}_j\right\|_2}\mathbf{K}_j, \tag{5}$$

where $\|\cdot\|_2$ represents L2 normalization, $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{t \times d}$. Although using the standard ReLU function can strictly satisfy non-negativity, this activation function may encounter the problem of zero gradient error, leading to unstable training processes. Additionally, the ReLU function sets negative elements to zero, leading to inevitable information loss issues, as each element should contribute to representation learning in the sequence. Therefore, we employ a variant of the ReLU series, the ELU function [27], to perform activation and control negative values. Ultimately, the student's global knowledge state $\mathbf{H}^E$ is learned through Eq. (3).

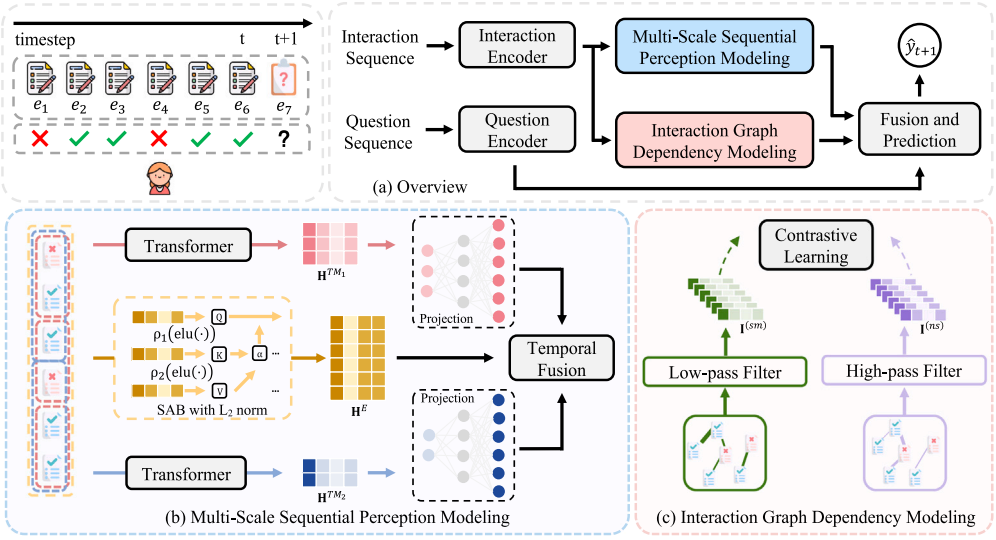


Fig. 3. The proposed MGIGKT framework. (a) is the overall workflow of the model. (b) is the Multi-Scale Sequential Perception Modeling module, dynamically modeling the student's knowledge state from a temporal perspective. (c) is the Interaction Graph Dependency Modeling module, modeling two types of interaction graphs based on similarity, and capturing the student's knowledge state from a static perspective.

Fine-grained modeling of students' knowledge states: Fine-grained modeling of students' historical learning records allows for more precise tracing of changes in their knowledge states over different learning periods, thereby providing a deeper understanding of their learning characteristics and behavior patterns [28]. To achieve this goal, we design a fine-grained attention network to more effectively capture knowledge state changes at specific granularities. Specifically, we introduce a time-aware encoder to generate knowledge state representations $\mathbf{H}^{TM_1}$ and $\mathbf{H}^{TM_2}$, where TM_1 and TM_2 represent the different temporal granularities. Since the steps for modeling knowledge states at both granularities are the same, we will omit the subscripts 1 and 2 in the following text and use $\mathbf{H}^{TM}$ directly. The formula is represented as follows:

$$\mathbf{H}^{TM'} = \{agg(\mathbf{I}_1^s, \ldots, \mathbf{I}_l^s), agg(\mathbf{I}_{l+1}^s, \ldots, \mathbf{I}_{2l}^s), \ldots, agg(\mathbf{I}_{t-l+1}^s, \ldots, \mathbf{I}_t^s)\}, \tag{6}$$

where the function $agg(\cdot)$ represents the aggregator operation of row vector addition. After obtaining representations capturing short-term knowledge states, a self-attention operation is applied to them to obtain granularity-specific knowledge state representations. The formula is represented as follows:

$$\mathbf{H}^{TM} = softmax(\frac{\mathbf{H}^{TM'}\mathbf{W}_{TM}^{Q}(\mathbf{H}^{TM'}\mathbf{W}_{TM}^{K})^{\top}}{\sqrt{d}})\,\mathbf{H}^{TM'}\mathbf{W}_{TM}^{V}, \tag{7}$$

where $\mathbf{W}_{TM}^{Q}, \mathbf{W}_{TM}^{K}, \mathbf{W}_{TM}^{V} \in \mathbb{R}^{t\times d}$ are the projection matrices mapping granularity-specific knowledge state representations to query, key and value. In our model, we configure transformer modules with two scales, ultimately generating three granularity-specific knowledge state representations: $\mathbf{H}^{E}$, $\mathbf{H}^{TM_1}$, $\mathbf{H}^{TM_2}$.

Finally, we integrate the three granularity-specific knowledge state representations into the same latent space using a multi-scale aggregation layer, as shown below:

$$\mathbf{H}^{S} = f_1\left(\mathbf{H}^{E} \parallel \mathbf{H}^{TM_1} \parallel \mathbf{H}^{TM_2}\right), \tag{8}$$

where $\parallel$ denotes matrix concatenation, and the function f_1 is a projection function that transforms the embedding matrix of dimension $3d$ into dimension d. Finally, we obtain the knowledge state matrix $\mathbf{H}^{S}$ of the student after being modeled by the sequence module.

3.2.3. Interaction graph dependency modeling (IGDM)

As previously mentioned, we believe that in real online education systems, there are latent correlations between the knowledge concepts involved in interactions. In addition to knowledge concept information, interaction data also includes information on whether students answered the exercises correctly. If we group interactions solely based on their association with knowledge concepts, we might overlook the correctness of the student's responses, leading to the loss of important answer outcome information [29]. To more accurately describe the complex relationships in students' historical interactions, we introduce an Interaction Graph Dependency Modeling module that can describe the relationships between interactions. Specifically, we construct two types of interaction graphs: one to represent the similarities between interactions and the other to describe their dissimilarities. Through these two graphs, we cannot only capture the direct correlations in students' historical interactions but also reveal the latent, deeper correlations. We apply Graph Neural Networks (GNNs) to these two graphs, performing low-pass and high-pass filtering operations. Specifically, we use the dissimilarity matrix to push apart the representations of dissimilar nodes, reducing interference from dissimilar neighbors, and the similarity matrix to bring the representations of similar nodes closer. This approach allows us to obtain distinctive representations of students' knowledge states. We further optimize these representations by employing a contrastive learning mechanism. This mechanism provides additional supervisory signals to the model by comparing knowledge state representations across different interaction graphs, guiding the model to generate more consistent and meaningful representations. Consequently, the model can predict students' future learning performance with greater accuracy, as it can now capture the subtle differences and complexities in the knowledge states.

Construction of Interaction Graphs: To model the relationships between students' historical interactions, we generate two types of interaction graphs. To improve computational efficiency, we use dot products to calculate the similarity a_{ij} between any two student historical interactions $\mathbf{I}_i$ and $\mathbf{I}_j$, and use this similarity value as the weight of the corresponding element in the adjacency matrix $\mathbf{A}^{(sm)}$ of the similarity graph. Then, we normalize the obtained similarity matrix row-wise to ensure that each weight value is between 0 and 1. Additionally, to ensure the simplicity and effectiveness of the graph structure, we set a similarity threshold, which is set to 0.5 in this paper. Only when the similarity between two nodes exceeds this threshold do we establish an edge between them. This approach helps to highlight nodes with strong correlations, thereby more accurately reflecting the structural information in the graph. The formula is represented as follows:

$$a_{ij}^{(sm)} = sigmoid(\mathbf{I}_i^{\top}\mathbf{I}_j), \tag{9}$$

$$\mathbf{A}^{(sm)} = sigmoid(\mathbf{I}^{\top}\mathbf{I}). \tag{10}$$

Next, based on the obtained similarity matrix $\mathbf{A}^{(sm)}$, we construct the adjacency weight matrix $\mathbf{A}^{(ns)}$ for the dissimilarity interaction graph.

$$\mathbf{A}^{(ns)} = \mathbf{J} - \mathbf{A}^{(sm)}, \tag{11}$$

where $\mathbf{J}$ represents a matrix where all elements are 1.

Dual-Channel Encoding: To comprehensively model the latent correlations between interactions and learn the static knowledge state of students from both similar and dissimilar interaction graphs, we design two encoders that perform low-pass and high-pass filtering on these two different views, respectively. This approach allows us to capture shared information among interactions based on similar edges, while filtering out irrelevant information between adjacent interactions based on dissimilar edges. Such a processing approach aids in more accurately identifying and leveraging the key features within interaction data, thereby enhancing the accuracy and efficiency of KT.

Using a low-pass graph filter can smooth features along the nodes [30], as low-pass filtering specifically captures the low-frequency information in the graph signal, thereby preserving the shared knowledge between similar nodes. Consequently, in the similarity interaction graph, we choose to use a simple low-pass GNN as the low-pass graph filter. This is represented as:

$$\mathbf{I}_0^{(sm)} = MLP_1(\mathbf{I}), \tag{12}$$

$$\mathbf{I}_l^{(sm)} = \mathbf{A}^{(sm)} \cdot \mathbf{I}_{l-1}^{(sm)} \cdot \mathbf{W}_{l-1}^{(sm)}, \tag{13}$$

where $\mathbf{W}_{l-1}^{(sm)} \in \mathbb{R}^{d\times d}$ is a learnable projection matrix.

Dissimilarity interaction graphs consist of edges that connect dissimilar nodes. In this view, applying a low-pass filter might smooth out the features of connected nodes, thus blurring the differences between dissimilar nodes and reducing node discriminability. To avoid this, we employ a high-pass graph filter (the inverse operation of a low-pass filter) to sharpen the features of nodes along the edges. The purpose of this is to distinguish between nodes that are different yet interconnected. The process of graph convolution on the dissimilarity interaction graph is consistent with that on the similarity interaction graph. The primary difference lies in replacing the adjacency weight matrix in the convolution operation with the dissimilarity adjacency weight matrix $\mathbf{A}^{(ns)}$ and different MLP_2. In this way, we ultimately obtain the knowledge state $\mathbf{I}^{(ns)}$ modeled based on the student's dissimilarity interaction graph.

By utilizing two dedicated encoders, we obtain two distinct sets of representations, each capturing low-frequency and high-frequency information, respectively. These representations reveal the characteristics of a student's knowledge state from two complementary perspectives. To integrate the information from these viewpoints, we concatenate the representations, thereby forming the final static representation of the student's knowledge state:

$$\mathbf{H}^G = f_2(\mathbf{I}^{(sm)} \parallel \mathbf{I}^{(ns)}), \tag{14}$$

where f_2 is a projection function that transforms the embedding matrix of dimension $2d$ into dimension d.

3.2.4. Fusion and prediction (FP)

Ultimately, we successfully construct two models of knowledge state: one, denoted as $\mathbf{H}^S$, which models learning patterns from multiple granularities, and the other, denoted as $\mathbf{H}^G$, which comprehensively simulates the latent correlations between interactions. To integrate these two perspectives of knowledge state, we employ an adaptive aggregation strategy. The aggregation process is as follows:

$$\alpha_t^S = \frac{\boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^S}{\boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^S + \boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^G}, \tag{15}$$

$$\alpha_t^G = \frac{\boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^G}{\boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^S + \boldsymbol{\alpha}^{\top} \cdot \mathbf{W}_{\alpha} \cdot \mathbf{H}_t^G}, \tag{16}$$

$$\mathbf{h_t} = \alpha_t^S \mathbf{H}_t^S \oplus \alpha_t^G \mathbf{H}_t^G, \tag{17}$$

$$\hat{y}_{t+1} = MLP(\mathbf{h}_t \oplus \mathbf{E}_{t+1}), \tag{18}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^d$ is learnable projection vectors, $\mathbf{W}_{\alpha} \in \mathbb{R}^{d\times d}$ is learnable projection matrices. α_t^S, α_t^G represent the importance of $\mathbf{H}_t^S$ and $\mathbf{H}_t^G$ at time t, used for the final aggregation of knowledge state.

Model Training: To train all the parameters in the model, we leverage the binary cross-entropy loss as a loss function between the predicted performance $\hat{y}_{t+1}$ and actual performance y_{t+1}. The following objective function is defined over training data:

$$L_p = -\sum_{t+1} \left(y_{t+1} log \hat{y}_{t+1} + \left(1 - y_{t+1}\right) log \left(1 - \hat{y}_{t+1}\right)\right). \tag{19}$$

To enhance the effectiveness of learning student knowledge state representations from two distinct graph structures, we incorporate the contrastive learning. This mechanism provides additional supervisory signals to the model by comparing the representations generated from these two different graph structures. This approach encourages the model to generate consistent semantic representations across both graph structures, thereby enhancing the quality of the representations and the overall performance of the model. Specifically, as shown in Fig. 4, for a given node $v_i^{(sm)}$, we first identify it as the anchor node in the similarity graph and compare it with $v_i^{(ns)}$ in the dissimilarity view to form a positive sample pair, aiming to make the representations of the positive sample pair closer. At the same time, we use other nodes in the dissimilarity view as negative sample pairs to pull the representations of these negative pairs further apart. A similar operation is also performed for each node in the dissimilarity graph. Through this dual-channel contrastive learning loss design, we can better distinguish the differences between nodes in the feature space, thereby enhancing the expressiveness and stability of the overall model.

$$L_c = -\frac{1}{n}\sum_{v_i\in\mathcal{V}}\left[\sum_{v_j\in\mathcal{N}_i}\frac{1}{2\left|\mathcal{N}_i\right|}\left(log\frac{e^{sim(\mathbf{I}_i^{(sm)},\mathbf{I}_j^{(ns)})/\tau_c}}{\sum_{v_k\in\mathcal{V}\setminus v_i} e^{sim(\mathbf{I}_i^{(sm)},\mathbf{I}_j^{(ns)})/\tau_c}}\right.\right.$$
$$\left.\left. + log\frac{e^{sim(\mathbf{I}_i^{(ns)},\mathbf{I}_j^{(sm)})/\tau_c}}{\sum_{v_k\in\mathcal{V}\setminus v_i} e^{sim(\mathbf{I}_i^{(ns)},\mathbf{I}_j^{(sm)})/\tau_c}}\right)\right], \tag{20}$$

where $sim(\cdot)$ employs the cosine similarity function, and τ_c is the temperature hyperparameter.

In this paper, our overall optimization objective is divided into two parts: the model prediction loss L_p and the contrastive learning loss of the interaction graph dependency module L_c. These two components together form our optimization objective, as shown in the specific formula below:

$$L(\theta) = L_p + \lambda L_c, \tag{21}$$

where θ represents all the parameters in this model. λ is used to control the impact of the contrastive loss on the model.

3.3. Time complexity analysis

MGIGKT employs a meticulously designed Multi-Scale Sequence Perception Modeling module, an Interaction Graph Dependency Modeling module, and a Fusion and Prediction module to accurately generate students' knowledge states. To understand this trade-off precisely, we conduct a time complexity analysis at a constant level, focusing solely on the two knowledge state modeling modules and the fusion and prediction module to provide a more accurate assessment of the overall time complexity. In the Multi-Scale Sequence Perception Modeling module, students' knowledge states are modeled both holistically and

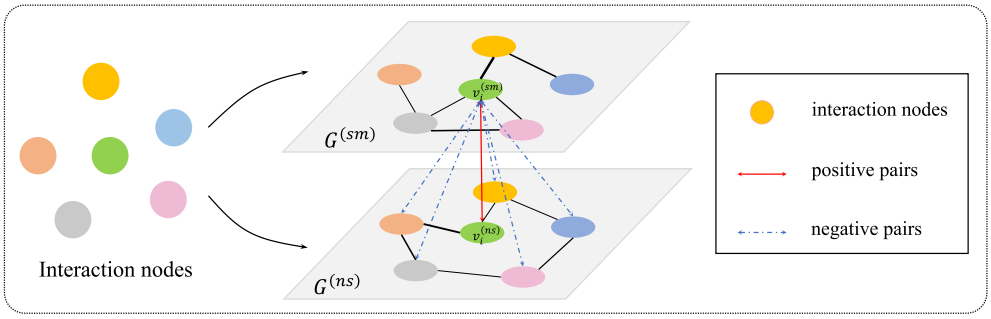


Fig. 4. The calculation process of contrastive learning loss. The solid red lines represent positive sample pairs, while the blue dashed lines represent negative sample pairs.

at different temporal granularities (Eqs. (3) to (8)). The associated time complexity is $O\left(t + \frac{t^2}{l^2_{TM_1}} + \frac{t^2}{l^2_{TM_2}}\right)d$, where t is the total length of the student's historical learning records, l_{TM_1} and l_{TM_2} represent the lengths of subsequences at different temporal granularities TM_1 and TM_2, and d is the size of the embedding dimension. Besides, the time complexity for the Interaction Graph Dependency Modeling module (Eqs. (9) to (14)) is $O(td^2 + t^2d)$. The time complexity of the contrastive learning loss included in this module is $O(|Ni|(nd + n^2))$, where $|Ni|$ denotes the number of neighbors for each node, and n represents the total number of nodes. For the Fusion and Prediction module (Eqs. (15) to (18)), the total complexity is $O(td^2)$. Finally, the overall time complexity of MGIGKT is $O((t + \frac{t^2}{l^2_{TM_1}} + \frac{t^2}{l^2_{TM_2}})d + td^2 + t^2d + |Ni|(nd + n^2))$.

3.4. Novelty analysis

Our method, MGIGKT, captures the multi-granularity learning patterns present in students' historical learning records and the latent correlations between interactions, offering a nuanced representation of students' knowledge states. This model revolves around two primary modeling processes: firstly, it engages in multi-scale sequential perception modeling on students' historical learning records, which showcase diverse temporal distribution characteristics across multiple time scales; secondly, it establishes two varieties of interaction graphs based on distinct response outcomes, facilitating interaction graph dependency modeling. Our work excels in the following pivotal aspects:

(1) Recognizing the diverse learning patterns exhibited by different students across varying temporal granularities, we have devised a methodology to capture these patterns through multiple time scales. This led to the development of a Multi-Scale Sequence Perception Modeling module aimed at simulating students' learning behaviors comprehensively at different granularities, thereby offering a more precise portrayal of their knowledge states. While recent studies like QKT [31] have focused on segmenting students' historical learning records based on quizzes, relying solely on a single granularity overlooks the intricacies inherent in the learning journey. In contrast, our approach delves into modeling students' knowledge states across diverse time granularities, promising an improved accuracy in temporal knowledge state representation.

(2) We introduce a pioneering Interaction Graph Dependency Modeling module that intricately captures the latent correlations existing between interactions. Traditional GCN-based approaches have primarily concentrated on relationships among students, interactions, and knowledge concepts. For example, SGKT [32] and HGKT [33] devise various graph structures to tackle knowledge tracing tasks. Nevertheless, these structures often fail to unveil the deeper interconnections within historical interactions, such as shared knowledge concepts across problems, and tend to overlook the significance of response correctness in interactions. In contrast, our method constructs interaction graphs based on both correct and incorrect responses, aiming to provide a more granular depiction of the interaction dynamics. It is worth noting that recent advances in node representation learning have validated the efficacy of modeling nodes from perspectives of similarity and dissimilarity. The GREET model [34] leverages dual-channel encoding to learn node representations via homophilic and heterophilic views, bolstered by an edge discriminator during training. Drawing inspiration from GREET, our approach not only introduces the construction of similarity and dissimilarity graphs for knowledge tracing tasks but also demonstrates utility for broader graph representation learning endeavors.

4. Experiments

Extensive experiments are conducted on four real-world datasets to evaluate the proposed model. The main objective of this section is to answer the following questions:

- **RQ1:** How does MGIGKT perform in tracing student knowledge states compared to existing KT methods?
- **RQ2:** What is the model's performance in terms of time complexity?
- **RQ3:** How effective are the key components designed in the model?
- **RQ4:** What is the impact of different hyperparameters on the model's performance?
- **RQ5:** Can MGIGKT learn meaningful student states?

To address these questions, we first provide a detailed description of the experimental settings, including dataset statistics, comparison baseline methods, and specific experimental settings. Then, we show the performance of the selected comparison baseline methods and our proposed model. Next, we conduct experiments to verify the time efficiency of the model. Following that, we conduct ablation experiments to study the effectiveness of each carefully designed component. Subsequently, we perform hyperparameter experiments to analyze the impact of key parameters on model performance and determine the optimal parameter settings. Finally, we visualize the changes in student knowledge states during the learning process through case studies and discuss the model's ability to trace knowledge states.

4.1. Datasets

This subsection introduces the four real-world public datasets used to evaluate the effectiveness of our designed MGIGKT. Table 2 shows the statistical information for all datasets. Below is a brief description of each dataset:

- **ASSIST2009** is collected from the ASSISTments online tutoring platform in the academic year of 2009–2010. We use the updated skill-build version of this dataset as it fixes data modeling issues and removes duplicated records [35]. Each exercise only involves one knowledge concept.
- **ASSIST2017** is collected from the ASSISTments Data Mining Competition in 2017.

Table 2
Statistical details of three datasets.

Dataset	ASSIST2009	ASSIST2017	EdNet	Eedi
#Student	4217	1709	5000	118,971
#Exercise	26,688	3162	10,795	27,613
#Knowledge Concept	195	102	188	282
#Interaction	346,860	911,613	225,625	15,867,849
#Avg. Length	82.25	873.79	–	133.38

- **EdNet** is a dataset collected by Santa over more than two years, encompassing all student-system interactions. EdNet offers four different abstract-level datasets, and in this paper, we utilize its simplest form, EdNet-KT1, consisting of students' exercise-solving records.
- **Eedi** consists of collected responses to mathematics problems from the British free primary and secondary school teaching platform Eedi between 2018 and 2020.

4.2. Baselines

To evaluate the performance, we compare MGIGKT with two groups of methods, i.e., Conventional KT methods and Graph-based KT methods. For each group, we select representative baselines and briefly introduce them below.

(1) Conventional KT methods

- **DKT** [8] introduces RNNs into the KT task for the first time, employing LSTM to trace the dynamic changes of students' knowledge states during the learning process.
- **DKVMN** [9] incorporates memory networks into the KT task, utilizing a static key matrix to store knowledge concepts and a dynamic value matrix to store and update students' knowledge states.
- **SAKT** [10] employs a self-attention mechanism to model students' knowledge states, considering the correlation between predicting exercises and past interactions to model students' knowledge states. Meanwhile, it can also address data sparsity issues.
- **MRTKT** [19] utilizes the Transformer architecture to model the various interaction relationships present in students' historical learning records, finely modeling the students' knowledge states.
- **DTransformer** [36] not only designs a proficiency modeling module from the exercise level to the knowledge concept level but also incorporates a contrastive learning mechanism to model the stability of knowledge states during the KT process.

(2) Graph-based KT methods

- **GKT** [11] constructs a graph structure where the nodes represent students, exercises, and knowledge concepts, and the edges represent the interactions between them. It uses GNNs to capture these interaction patterns and infer the students' knowledge states.
- **DGEKT** [12] establishes hypergraphs based on knowledge concepts and constructs directed transition graphs according to students' learning sequence time information. It then employs a knowledge distillation framework to integrate the modeling of student knowledge states based on the two types of graphs.

4.3. Experimental settings

To evaluate our model, we compare its performance with all baseline methods on all datasets mentioned earlier. For each dataset, we allocate 80% of students to the training set, 10% to the validation set, and 10% to the test set. The dimension of knowledge states is set to 128, and the dimension of projection matrices in the attention mechanism is set to 128. We remove students with fewer than 10 historical learning records because too few historical learning records are meaningless. The optimizer used is Adam, with a learning rate of 0.00001. We implement our model using the PyTorch framework. All models are tuned to achieve the best performance for a fair comparison. We use ACC (Accuracy) and AUC (Area Under the ROC Curve), commonly used in the KT task, as evaluation metrics for the model.

Table 3
Comparison of all baseline methods and MGIGKT. The best performance results are in bold, and the second are underlined.

	ASSIST2009		ASSIST2017		EdNet		Eedi	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
DKT	0.785	0.763	0.726	0.691	0.703	0.667	0.765	0.720
DKVMN	0.789	0.746	0.707	0.711	0.681	0.658	0.758	0.716
SAKT	0.799	0.765	0.657	0.702	0.751	0.747	0.819	0.728
MRTKT	0.822	0.784	–	–	0.775	0.732	0.826	0.757
DTransformer	0.815	0.763	0.751	0.708	0.693	0.679	0.739	0.712
GKT	0.740	0.696	0.692	0.638	0.679	0.613	0.701	0.692
DGEKT	0.766	0.744	0.776	0.757	0.701	0.688	–	–
MGIGKT	**0.834**	**0.798**	**0.828**	**0.797**	**0.805**	**0.769**	**0.843**	**0.801**

4.4. Performance comparison(RQ1)

Table 3 presents a performance comparison between baseline methods and our proposed method across different datasets. The observed results are as follows:

(1) Among conventional KT methods, MRTKT performs the best. This method utilizes an attention mechanism that considers the different relationships between students' historical interactions. Additionally, SAKT achieves suboptimal results due to it uses an attention mechanism to model the correlations between interactions, considering the underlying relationships between them. DTransformer also performs well, as it constructs students' knowledge states from the exercise level to the knowledge concept level and uses a training algorithm based on contrastive learning. DKT performs well, indicating that simple models can sometimes achieve better performance. DKVMN often shows poorer performance because, although it can model knowledge concepts in fine granularity over time, in practical applications, the relationships and dependencies between knowledge concepts contained in interactions can be more complex. Simple key–value pairs may not sufficiently capture these relationships. This further validates the importance of considering the correlations between interactions in KT tasks.

(2) In graph-based KT methods, DGEKT performs the best because it considers not only the relationships between exercises and knowledge concepts but also the transition patterns of interactions. This further validates the importance of considering various interaction relationships in modeling student knowledge states.

(3) Graph-based KT methods generally do not perform as well as conventional KT methods because the KT task is inherently a time series forecasting problem. Modeling students' knowledge states from a graph perspective serves as an auxiliary approach. However, comprehensively modeling the latent correlations between interactions from a graph perspective can greatly enhance the performance of KT models.

(4) Overall, the performance of MGIGKT is superior to all baseline methods in terms of both AUC and ACC metrics, with a certain degree of improvement compared to the best baseline methods. This indicates that considering modeling learning patterns at multiple granularities and comprehensively modeling the latent correlations between interactions are highly beneficial for enhancing the performance of KT models.

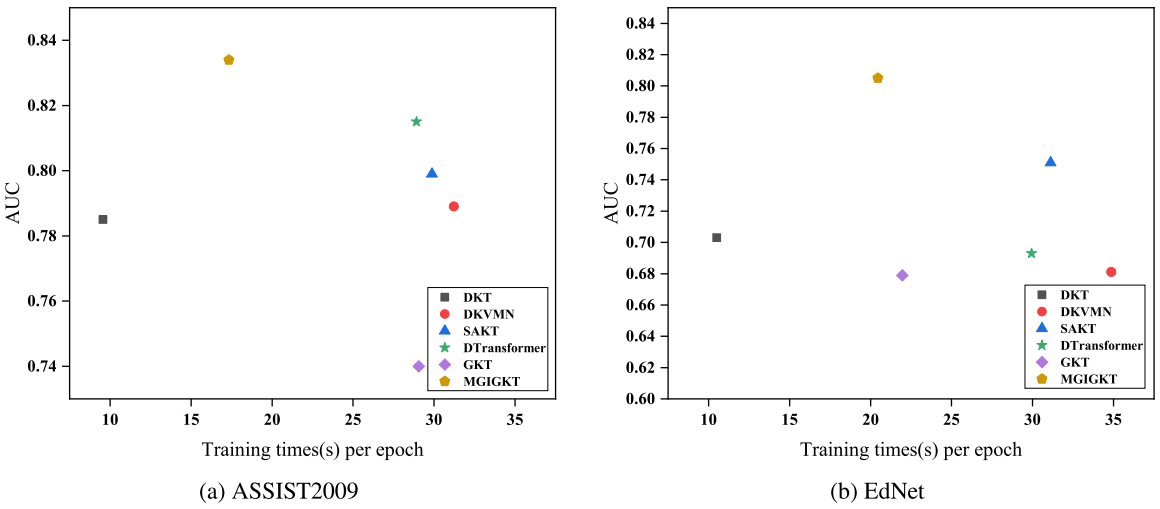


Fig. 5. Model comparison of the time efficiency on two datasets.

4.5. Comparison with the efficiency of various baselines(RQ2)

To address RQ2, this section presents a comparative analysis of our model's runtime and performance against five state-of-the-art knowledge tracing methods (namely, DKT, DKVMN, SAKT, DTransformer, and GKT). All models are evaluated under identical hardware conditions. Fig. 5 illustrates a scatter plot of experimental results on two knowledge tracing datasets of different scales, where the y-axis represents model AUC and the x-axis represents runtime per epoch. Models closer to the upper left corner exhibit superior performance.

As shown in Fig. 5, our proposed method consistently achieves either the best or second-best performance across both datasets, while maintaining an acceptable runtime, outperforming other baseline models. From a performance perspective, the superiority of our approach is mainly attributable to the inclusion of two core modules: a multi-scale sequence perception modeling module and an interaction graph dependency modeling module. These modules perform in-depth modeling of students' knowledge states from dynamic and static perspectives, respectively. On one hand, this design captures students' learning patterns at multiple time granularities more comprehensively; on the other hand, it models the underlying dependencies between interactions, significantly enhancing the model's accuracy in the knowledge tracing task. In terms of efficiency, although our method involves higher computational complexity due to the use of multiple Transformer modules, resulting in a mid-range runtime, we mitigate the high time complexity of Transformers in modeling overall student knowledge states by introducing an L2 regularization strategy. Despite requiring more computation time than the fastest method, DKT, our model's substantial performance improvement keeps its runtime within an acceptable range.

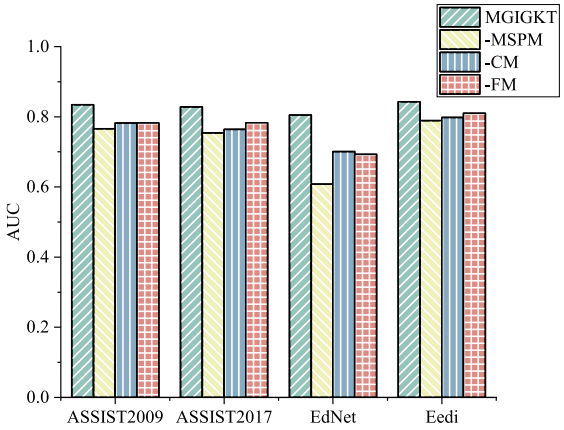
4.6. Ablation study(RQ3)

In this subsection, to justify the rationality of key components and contrastive learning loss in MGIGKT, we conduct experiments on several variants of MGIGKT and the results are shown in Fig. 6.

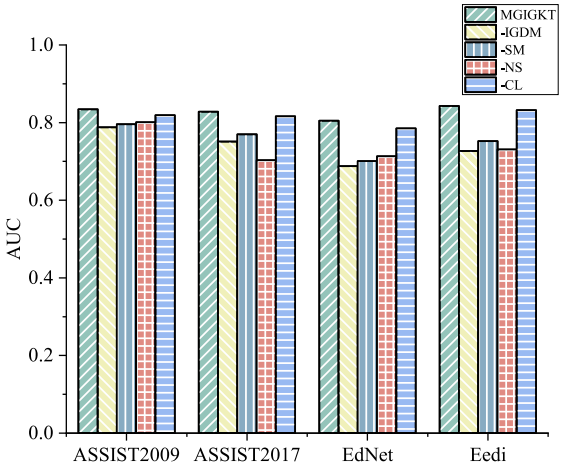
Effect of Multi-Scale Sequential Perception Modeling module. To investigate whether the Multi-Scale Sequential Perception Modeling (MSPM) module can effectively model students' learning patterns at multiple granularities, we design three variants: MGIGKT(w/o)-MSPM, MGIGKT(w/o)-CM, and MGIGKT(w/o)-FM. The first variant completely removes the MSPM module, the second variant removes the coarse-grained modeling of students' knowledge states using L2 regularization, and the third variant removes the fine-grained modeling of students' knowledge states. By analyzing the results in Fig. 6(a), we observe that compared to our MGIGKT model, these three variants show varying degrees of performance decline on the ASSIST2017 dataset. This validates two key findings: (1) The MSPM module can effectively model students' learning patterns at multiple granularities and capture sufficient key feature information; (2) There is a mutually supportive and complementary relationship between the coarse-grained and fine-grained modeling modules, which is crucial for comprehensively modeling students' knowledge states and improving the performance of KT models.

Effect of Interaction Graph Dependency Modeling module. To validate the effectiveness of the Interaction Graph Dependency Modeling (IGDM) module, we introduce three variants: MGIGKT(w/o)-IGDM, MGIGKT(w/o)-SM, and MGIGKT(w/o)-NS. These variants represent the removal of the IGDM module, the similarity interaction graph modeling module, and the dissimilarity interaction graph modeling module from the original model, respectively. The results in Fig. 6(b) indicate that these three variants show performance declines across all four datasets. This finding strongly supports the importance of the IGDM module, which is specifically designed to comprehensively capture the latent correlations present in students' historical interaction sequences. The IGDM module is crucial for more accurately predicting changes in students' knowledge states and enhancing the performance of Knowledge Tracing (KT) models. Further analysis reveals that removing either the similarity or dissimilarity interaction graph affects model performance. This is because the similarity and dissimilarity graphs capture the similarities and differences between student interactions, respectively, complementing each other and collectively enhancing the model's performance.

Furthermore, we make two additional observations: **Obs1:** When analyzing the ASSIST2017 dataset, we find that the impact on model performance from removing the entire IGDM module is less noticeable compared to removing only a portion of it. Our investigation suggests that this phenomenon may be attributed to the substantial volume of interactions present in this dataset, leading to lower similarity levels between interactions in contrast to other datasets. This decreased similarity influences the construction of the similarity graph, resulting in inadequate similarity between nodes and consequently generating lower-quality homophilic graphs. Hence, improving the quality of these homophilic graphs emerges as a crucial avenue for future research. **Obs2:** Across the ASSIST2017 and Eedi datasets, the removal of the high-pass filter from the heterophilic graph has a more pronounced impact. This observation indicates that the significance of the similarity and dissimilarity graphs may vary depending on the data distribution within the datasets. These insights further emphasize the importance of our method's dual-channel encoding strategy, which concurrently



(a) Ablation study AUC performance comparison of the MSPM module on four real-world datasets.



(b) Ablation study AUC performance comparison of the IGDM module on four real-world datasets.

Fig. 6. Performance comparison between our method and variants.

considers both similarities and dissimilarities between interactions. This underscores the effectiveness and necessity of our approach.

In summary, retaining the interaction graph modules is essential for constructing an effective knowledge tracing model, as they are crucial for more accurately predicting changes in students' knowledge states and improving KT model performance.

Effect of Contrastive Learning Loss: To investigate the role of contrastive learning loss in improving the accuracy of predicting students' knowledge states, we design a variant, MGIGKT(w/o)-CL, where the contrastive learning loss is removed from the total loss. The experimental results, shown in Fig. 6(b), combined with the ablation study results on the IGDM module, reveal that MGIGKT(w/o)-IGDM performs the worst, indicating that completely removing the IGDM module has the most significant impact on model performance. The two variants that retain single-channel encoding still include contrastive learning loss—MGIGKT(w/o)-SM and MGIGKT(w/o)-NS—outperform MGIGKT(w/o)-IGDM. This suggests that even without dual-channel encoding, contrastive learning loss can still enhance model effectiveness to some extent. Furthermore, the variant MGIGKT(w/o)-CL, which lacks contrastive learning loss, performs better than the single-channel variants MGIGKT(w/o)-SM and MGIGKT(w/o)-NS, highlighting the critical role of dual-channel encoding. Nevertheless, the complete model that combines dual-channel encoding and contrastive learning loss achieves the best performance. This is because dual-channel encoding can capture shared information between interactions using similar edges and filter out irrelevant information between neighboring interactions through dissimilar edges. Meanwhile, contrastive learning loss acts as a constraint mechanism, encouraging the model to generate consistent semantic representations across the two graph structures, thereby enhancing representation quality and overall model performance.

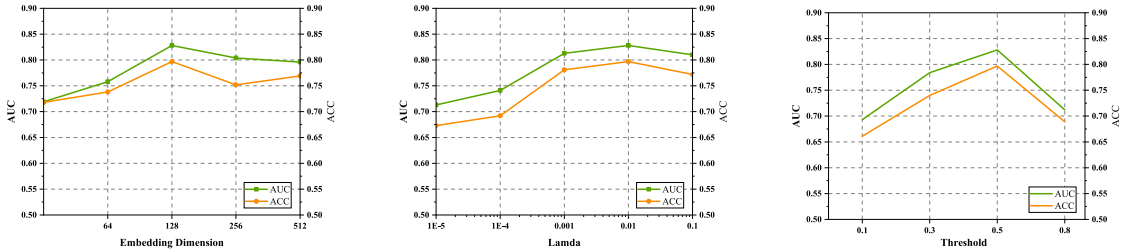
4.7. Hyperparameter analysis(RQ4)

In this subsection, we investigate the impact of different settings for several key hyperparameters in MGIGKT, including the embedding dimension d, the loss weight λ, which is used to control the impact of the contrastive loss on the model, and the threshold w. The results are shown in Fig. 7.

Embedding Dimension: The embedding dimension d determines the amount of information the model can capture. As shown in Fig. 7(a), we present the hyperparameter analysis results for the embedding dimension on the ASSIST2017 dataset. It can be observed that as the dimension gradually increases, the model's performance initially improves but then levels off. Ultimately, the performance peaks at $d = 128$. Compared to the optimal d, smaller or larger d values lead to a decrease in performance. This is because smaller d values cannot capture enough feature information, limiting the model's expressiveness and performance. On the other hand, larger d values may introduce too much noise and redundant information, causing the model to overfit and thus affecting its generalization capability and performance.

Contrastive Learning Loss Weight: The loss weight λ plays a crucial role in determining the importance of contrastive loss on the model's performance. As shown in Fig. 7(b), we present the hyperparameter analysis results for the loss weight on the ASSIST2017 dataset. The model's performance gradually improves with increasing loss weight in the range of 0.00001 to 0.1, peaking at $\lambda = 0.01$. After this point, the performance slightly decreases as the loss weight exceeds this value. This change may be because smaller loss weights fail to sufficiently emphasize the contribution of contrastive loss, reducing the model's efficiency in learning and distinguishing different features. On the other hand, excessively large loss weights may overly emphasize contrastive loss, causing the model to focus too much on contrastive learning while neglecting the learning of other critical features, negatively impacting overall performance. This result indicates that the loss associated with prediction contributes more significantly to model training, while the contrastive loss serves an auxiliary role, helping the model better learn and differentiate features.

Threshold: In the IGDM module, when constructing the similarity graph, we filter out edges with a similarity below the threshold w, setting their similarity to zero in order to reduce noise. To determine the optimal value of w, we conduct hyperparameter analysis experiments to investigate the impact of threshold selection on model accuracy and efficiency during matrix construction, with results shown in Fig. 7(c). It ca be observed that when w is set too low, the similarity graph (from the perspective of homophily) becomes contaminated with a large number of heterophilic edges, significantly degrading the quality of representations learned through low-pass filtering. Moreover, the number of heterophilic edges in the dissimilarity graph will also decrease, further impairing learning effectiveness. Conversely, when w is set too high, the dissimilarity graph (from the perspective of heterophily) may include many homophilic edges, resulting in poorer quality of representations learned through high-pass filtering, while the number of homophilic edges in the similarity graph decreases, negatively affecting representation quality as well. Therefore, we select 0.5 as a reasonable initial threshold and validate its effectiveness on the ASSIST2017 dataset. However, since different datasets may have



(a) Hyperparameter analysis of embedding dimension.

(b) Hyperparameter analysis of λ.

(c) Hyperparameter analysis of w.

Fig. 7. Hyper-parameter analysis of MGIGKT.

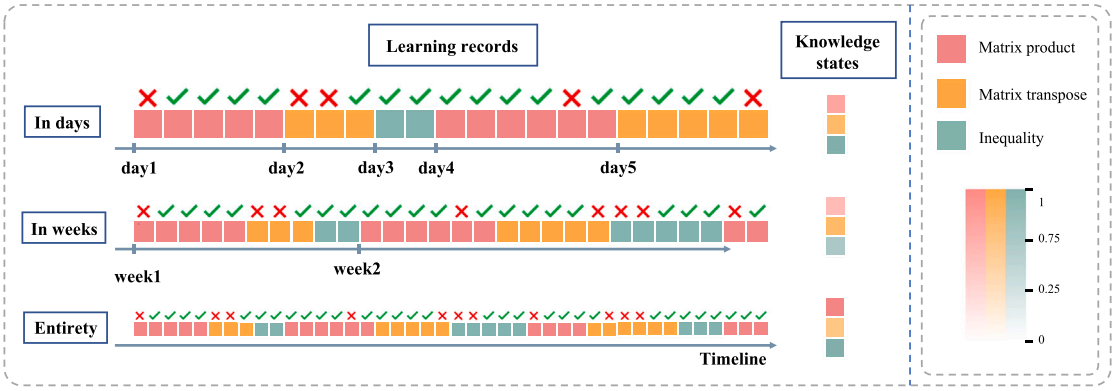


Fig. 8. Visualization of the fine-grained modeling process of students' learning patterns. The left part shows the students' historical learning records at different time granularities (day, week, and overall), with the knowledge state after completing the last exercise at each time granularity marked. The right part is the legend, which explains the correspondence between different colors and knowledge concepts in the left chart, as well as how the mastery of knowledge concepts is represented by changes in color intensity.

varying requirements for the threshold, how to adaptively determine this hyperparameter remains an area for further research.

4.8. Case studies(RQ5)

To gain a deeper understanding of how the MGIGKT model tracks students' evolving knowledge states, this section presents a visual analysis of the knowledge tracing process from two different perspectives. Specifically, we select the historical learning records of the student with ID 23 from the ASSIST2009 dataset and illustrate the changes in the student's knowledge state under two key modules—the Multi-Scale Sequence Perception Modeling module and the Interaction Graph Dependency Modeling module. In the Multi-Scale Sequence Perception Modeling module, we observe the student's knowledge state changes at different granularities. In the Interaction Graph Dependency Modeling module, we display the constructed interaction graph and the student's knowledge state at each moment.

Case study of modeling learning patterns at multiple granularities: To evaluate the KT capability of the MGIGKT model, we employ a multi-granularity analysis approach to visualize changes in students' knowledge states. Fig. 8 illustrates the learning activities of the student at daily, weekly, and overall granularities, and provides the final knowledge states at different granularities. By observing her daily learning activities, we are able to trace her mastery of knowledge concepts matrix product, matrix transpose, and inequality. For instance, on Monday, her performance in matrix product improved, while on Tuesday, due to mistakes in matrix transpose-related questions, her mastery level decreased. By Wednesday, she answers all questions related to inequality correctly, thus improving her mastery of inequality. Further analysis of the student's weekly learning activities provides an overall view of her grasp of the three knowledge concepts. In the first week, her accuracy in matrix product and inequality is high, whereas it is low in matrix transpose, indicating a need for more practice in matrix transpose. By the second week, her accuracy in matrix product and matrix transpose have improved, but it decreases in inequality, highlighting her weaknesses in inequality and suggesting a need for more focused practice on this knowledge concept. A comprehensive analysis of all her learning activities gives us an overall view of her performance in the three knowledge concepts. As shown in the figure, her performance in matrix product is the best, her performance in matrix transpose needed improvement, and she has a good grasp of inequality.

Case study of comprehensively modeling the latent correlations between interactions: We provide the historical learning records of the student, which include six interactions corresponding to three knowledge concepts with sequential relationships. We present the construction of similar and dissimilar interaction graphs from an interaction graph dependency perspective. It can be observed that when there are sequential relationships between knowledge concepts, the similarity between interactions also increases to a certain extent. When the exercises involve the same knowledge concepts but with different answers, the similarity between interactions decreases, while dissimilarity increases. As shown in Fig. 9, when a student correctly answers exercises related to the red knowledge concept, the proficiency in the red knowledge concept increases, whereas it decreases when they answer exercises related to the red knowledge concept incorrectly. When correctly answering exercises related to the yellow knowledge concept, the proficiency of the prerequisite knowledge concepts also increases simultaneously.

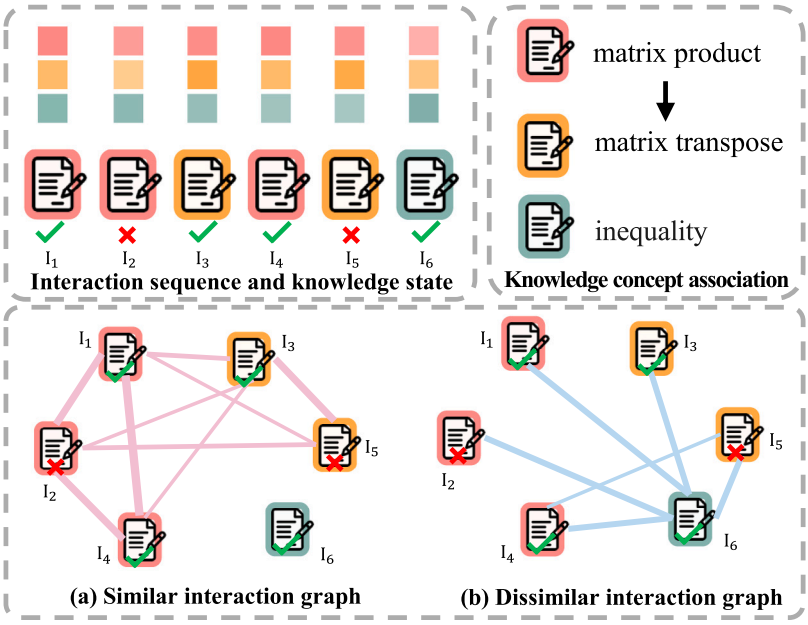


Fig. 9. Visualization of the KT process, where the upper part of the figure presents the student's historical learning records, and the lower part depicts the generated interaction graphs.

In summary, there are two key observations:

- By combining the results from fine-grained, mid-grained, and overall levels of modeling, we can gain a more comprehensive understanding of the student's learning state and provide personalized learning suggestions accordingly.
- Considering the correlations of interactions between students is significantly beneficial for accurately tracing and understanding their knowledge states, which not only aids the model in deeply uncovering the underlying connections within historical interaction data but also effectively captures the dynamic changes in a student's knowledge state when answering exercises involving potentially related knowledge concepts.

5. Conclusion

In this work, we propose a KT model named MGIGKT, which models students' knowledge states from dynamic, time-based perspectives and static, interaction-graph-based perspectives to predict their future performance. We first finely partition students' historical learning records, modeling their knowledge states granularly in the Multi-Scale Sequential Perception Modeling module to explore learning patterns at multiple granularities. Simultaneously, we statically model students' knowledge states from two constructed interaction views in the Interaction Graph Dependency Modeling module to capture the latent correlations in student interactions comprehensively. MGIGKT captures different types of contextual information in student learning records, enhancing its modeling capability. Experimental validation on four datasets confirms the effectiveness of our proposed model.

CRediT authorship contribution statement

Jing Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Formal analysis, Data curation. **Huifang Ma:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Funding acquisition, Data curation. **Mengyuan Zhang:** Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Data curation. **Lei Zhang:** Writing – review & editing, Resources, Project administration, Methodology, Investigation. **Liang Chang:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data used in this paper are all publicly available.

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